Prebuilt

Every printer learns from every other. No data leaves.

Every printer is wrong in its own way

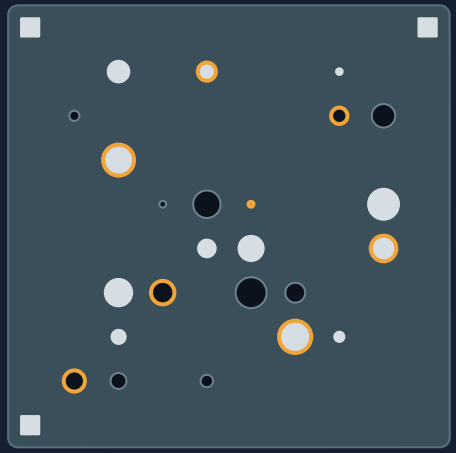
Holes print up to 250 µm small and pins up to 195 µm large (desktop SLA, CMM; Tehnički vjesnik 2026).

In medicine, that decides the fit

Crowns, surgical guides and microfluidic chips must fit first time. A crown margin allows ~120 µm.

And labs can't pool what they learn

Patient scans and process recipes can't leave, so every lab re-learns its printer from scratch.



The test plate: 24 pins and holes. Print it, measure it, know your printer.

HOW IT WORKS

Learn locally, share three numbers

1 Print a test plate

24 pins and holes, measured by micrometer or flatbed scanner.

2 Learn your printer

A physics model of your machine, fit locally on your own data.

3 Share three numbers

Mean and spread of 3 parameters every printer shares, logged on a receipt.

Never leaves the lab: patient scans, part geometry, measurements, machine settings. A replayable audit checks every crossing.

What's new: correcting printer error (Huang et al. 2015) and learning across printers (Wei et al. 2023) are proven, but pool the data. Prebuilt does it privately.

RESULTS · SIMULATION

Parts that fit, and new labs get there faster

9 µm

size error from a corrected design, against 132 µm drawn as-is.

35%

lower error for data-poor labs when eight share (won 20 of 20 fleets).

12%

lower error with just two labs, matching pooling all raw data.

0 rounds

lost to a lying lab: 180 of 180 lies caught, 0 honest notes dropped.

Registered first

Claims committed, with a hash, before the first print.

Negative control

Printers that share no physics gain nothing: the gain is real.

Real printer

A Form Studio 1 pin: 1.44 mm for 1.50, inside the ±0.08 mm spec.

WHO IT'S FOR

For anyone whose precision is a trade secret

NOW · MEDICAL AND BIO

- Hospital 3D-printing labs
- Dental labs
- Bio and microfluidics labs

NEXT · HIGH-PRECISION INDUSTRY

- Chip makers, GPU packaging
- Defense and aerospace
- MEMS and nanotech

Value grows with every member: 11% → 28% → 35% lower error at 2 / 4 / 8 labs. From microns today to nanometres next.

Print anywhere, trust the part.

github.com/matte-realize/prebuilt